

Kalhara Tennakoon

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on industry experience in software development, cloud-native systems, and automation. Experienced in designing scalable applications, building reliable deployment workflows, and implementing security-focused engineering practices through DevOps and Infrastructure as Code (IaC). Skilled in Kubernetes, Docker, CI/CD automation, backend development, and cloud platforms, including Azure and AWS. Passionate about DevSecOps, MLOps, AI infrastructure, cloud-native technologies, and building impactful software solutions through continuous learning and technical collaboration.

TECHNICAL EXPERTISE

Programming: Python, Java, Golang, JavaScript, C, Bash/Shell Scripting

Cloud & DevOps: Docker, Kubernetes, Tekton, ArgoCD, GitHub Actions, Azure DevOps, CI/CD, Infrastructure as Code (IaC)

Cloud Platforms: Microsoft Azure, AWS, Google Cloud Platform (GCP), AKS, GKE, Rancher

Web & Databases: React, Node.js, Flask, REST APIs, PostgreSQL, MySQL, MongoDB

Data & AI: Scikit-learn, pgvector, Ollama, Power BI, Jupyter Notebook, Data Cleaning, Exploratory Data Analysis (EDA)

Tools & Practices: Git, GitHub, Agile, Scrum, Jira, Technical Documentation

WORK EXPERIENCE

DevOps Engineering Intern

IFS R&D International (Pvt) Ltd, Colombo, Sri Lanka | Jul 2024 – Oct 2024

- Automated Kubernetes-native CI/CD workflows using Tekton, streamlining deployment processes for critical services.
- Integrated Behave automated testing into Tekton CI/CD pipelines, reducing manual debugging effort by ~30% and improving deployment reliability.
- Optimized pipeline configurations and resolved recurring build issues to improve deployment stability and engineering efficiency.
- Conducted a knowledge-sharing session on testing best practices to support team collaboration and code quality improvements.

Software Engineering Intern

IFS R&D International (Pvt) Ltd, Colombo, Sri Lanka | Jul 2020 – Dec 2020

- Developed a cloud security scanning application using Docker and Kubernetes to improve container platform security and vulnerability assessment workflows.
- Integrated security scanning tools into CI/CD pipelines, improving vulnerability detection across deployment stages.
- Diagnosed deployment and runtime issues while authoring technical documentation to support system reliability and onboarding processes.

EDUCATION

BSc (Hons) Computer Science (Software Engineering) – First Class Honours

Kingston University, London, United Kingdom | Graduation: September 2026

- Final Year Project: VetCare Pro – Smart Veterinary Clinic Management System | Grade: A (76%)

PROJECTS

VetCare Pro – Smart Veterinary Clinic Management System

React, Node.js, PostgreSQL, pgvector, Python, Flask, Scikit-learn, Ollama, SwiftUI

- Developed a full-stack veterinary clinic management platform supporting appointment scheduling, medical records, billing, and inventory workflows, with role-based access control, a pet owner web portal, and a SwiftUI companion app.
- Built a retrieval-augmented generation (RAG) assistant using locally hosted language models and pgvector semantic search, enabling staff to query clinic records conversationally while keeping patient data within the system.
- Implemented intelligent query routing that directs counting and aggregate questions to exact SQL handlers instead of semantic retrieval, improving factual accuracy for numerical queries and reducing unreliable responses.
- Designed human-in-the-loop controls for assistant-initiated database changes, requiring explicit user confirmation before committing proposed actions; integrated ML services for disease prediction, sales forecasting, and inventory demand forecasting.

Predicting Course Difficulty from Student Evaluation Responses

Python, Scikit-learn, Pandas, Matplotlib, SPSS, Machine Learning

- Developed supervised machine learning models to predict perceived course difficulty using student evaluation datasets and statistical analysis techniques.
- Performed data preprocessing, exploratory data analysis, feature engineering, and classification model evaluation using Python-based data science tools.
- Identified instructor-related attributes as significant predictors through statistical validation and model interpretation.

Kubernetes Cluster & Container Image Security Scanner

Kubernetes, Docker, Golang, ArgoCD, Rancher, AKS, GKE, Azure DevOps

- Developed a containerized security scanning solution for Kubernetes environments to detect vulnerabilities in container images and cluster configurations.
- Integrated automated security scanning workflows into CI/CD pipelines to strengthen secure deployment practices and vulnerability detection coverage.
- Utilized cloud-native platforms and GitOps tooling to support scalable, secure, and reliable Kubernetes operations.

CERTIFICATIONS

Cloud, DevOps & Platform Engineering

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Introduction to Kubernetes – The Linux Foundation
- GitHub Foundations
- Containers & Kubernetes Essentials – IBM
- Introduction to GitOps

Artificial Intelligence & Data

- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Data Fundamentals (DP-900)
- Google AI Professional Certificate

ACHIEVEMENTS

- **Two-Time Dan Kohn Scholarship Recipient – KubeCon + CloudNativeCon Europe (2022 & 2025)**
 - Awarded the highly competitive scholarship recognizing contributions to the cloud-native ecosystem.
- **First Runner-Up – Hack:Bit (2019)**
 - Secured 2nd place in a competition organized in collaboration with Microsoft and Sarvodaya Fusion.
- **Top 15 Finalist – HaXmas Hackathon (2018)**
 - Selected among 75+ competing teams in a national-level hackathon.
- **Finalist – Cisco Packet Riders (2018)**
- **Cisco NetRiders (2017)**
 - Ranked 4th in Sri Lanka and 118th in the Asia Pacific & Japan region.

TECHNICAL WRITING & COMMUNITY ENGAGEMENT

- **Technical Writer – Medium | Dec 2021 – Present**
 - Authored 50+ technical articles on DevOps, cloud-native technologies, and software engineering topics.
 - Generated 150K+ views on DevOps and cloud-native content, demonstrating strong technical communication and community engagement.
- **Volunteer – STEMUp Educational Foundation | Aug 2018 – Dec 2020**
 - Contributed to initiatives promoting STEM education and digital literacy among students and youth communities.